

Q1) 5 interesting concepts related to vision:

1. Human Brain Actively Erases Your Nose:

Because of the way in which eyes are positioned on the face, the nose is constantly blocking a portion of the field of vision. However, the brain realizes that staring at one's own nose isn't useful, so it actively filters it out of conscious awareness. We only actually "see" it when we try to remember it's there and focus on it.

2. The night vision is entirely Black and White:

The human eye relies on two main types of light sensors: "cones" (which detect color and fine detail) and "rods" (which detect basic shapes in low light). Because the color-detecting cones need a massive amount of light to function, they completely shut down in the dark. That means when people are actively trying to navigate in the dark without any light, the entire world appears as different shades of black and white.

3. Our brain constantly guesses colors:

The visual system doesn't just passively record colors, but rather it actively guesses them based on the lighting around. For example: If the brain perceives that an object is sitting in a blue shadow, it will mentally "subtract" the blue light to show what it thinks the object's true color is. This constant, automatic visual editing is exactly why famous optical illusions (like the viral "black/blue vs. white/gold dress") manage to break our brains.

4. Eagles Have a Biological "Zoom Lens":

While humans have just one small area of sharp, high-definition focus in the center of our eyes, birds of prey like eagles and hawks have two. Furthermore, the muscles in an eagle's eye can change the curvature of

their lenses to magnify their field of view. This biological zoom allows them to spot a target as small as a rabbit from over two kilometers up in the sky.

5. Emotions also control the pupils and not just light:

Pupils automatically expand in the dark and shrink in the light to accommodate only a certain amount of light to pass through. But they are also wired directly to the autonomic nervous system. When we experience sudden fear, or even when you are just concentrating very hard on a difficult problem, the pupils can subconsciously dilate by as much as 45%.

Q2) Different types of healthcare reports and medical imaging reports:

Healthcare reports are formal documents used to record, communicate, and analyze patient care and hospital operations. They broadly fall into five distinct categories:

- **Clinical Reports:** The core of patient documentation, including documents like Progress Notes, Discharge Summaries, and Nursing Notes. These reports enable tracking a patient's health journey and actively support informed clinical decision-making.
- **Diagnostic Reports:** These provide objective, measurable data required for a diagnosis. This category includes laboratory, pathology, EKG, and medical imaging reports, which are among the most complex documents in healthcare.
- **Administrative Reports:** They are created when anyone tries to avail the healthcare system. These reports support operational management. While often collected for billing, the data is also vital for analyzing the cost-effectiveness and overall efficiency of healthcare delivery.

- **Financial Reports:** These translate clinical services into billing data. By analyzing this information, practitioners and institutions can evaluate resource allocation, monitor staff expenses, and identify areas where excessive money is spent.
- **Regulatory and Public Health Reports:** Compulsory external submissions such as basic statistical reports and cost reports, which are sent to government bodies or public health authorities to ensure patient safety and quality compliance standards are met.

Medical Imaging Reports:

After analyzing scans like X-rays, MRIs, CTs, or ultrasounds, a radiologist produces a specialized medical imaging report. This essential document acts as the main line of communication between the radiologist and the patient's doctor, translating complicated visual findings into clinical insights, which will help in further diagnosis.

Radiology reports are very important in legal practice. A detailed report can distinguish between medical malpractice and an unavoidable error. Accuracy and precision play an important role as an absence of a document means that the finding was not observed.